Whose Perception of Geopolitical Risk Is Priced in Defence Equities?

Evidence from the Russia–Ukraine War

Khrystyna Kateryna Valenia

August 2026

Abstract

Standard measures of geopolitical risk are constructed from English-language newspapers, which makes the editorial vantage point of those measures an implicit modelling assumption rather than a tested one. This thesis asks whose perception of geopolitical risk is priced in defence equities. Using 4,027 days of multilingual news from GDELT's translingual archive, classified by the national identity of the *publisher* rather than by the subject of the article, it tests whether Ukrainian, Russian state and Russian independent coverage of the Russia–Ukraine war explains defence-equity returns conditional on Western coverage. It does not. Local-language perception adds no explanatory power in coverage volume, in tone, or in the anticipated-versus-realised structure of coverage, across two asset classes — defence equities and European natural gas — and one non-market outcome, realised conflict escalation. The design is nonetheless sensitive: the Western block is detected under multiple-testing correction, and local coverage is itself detected in seven of 31 specifications when credited to its publication day, falling to one when lagged by the single day required for tradeability. A power simulation on 1,855 out-of-sample days detects a true out-of-sample R^2 of 0.5% in 83% of samples, so effect sizes the predictability literature treats as economically meaningful are excluded rather than merely unobserved. Descriptively, Russian state media's conflict tone did not change at the invasion (+0.02) while Ukrainian media's fell by 1.66 points, indicating that the signal which moved most sharply at the defining event of the

sample is not the one defence-equity prices track.

Contents

1 Introduction

The Russia–Ukraine war is the most intensively covered armed conflict in modern media history, and it produced the sharpest repricing of defence equities in decades. European defence equities rose 21.6% in the five weeks following the 24 February 2022 invasion, against 7.0% for the global sector index and 9.1% for its U.S. counterpart; the sector then re-rated far more dramatically in the first half of 2025 (Europe +63.7%) as European governments announced rearmament budgets. For an investor trying to understand that repricing, the natural question is: what information is driving it? The standard answer in the geopolitical-risk literature is "what newspapers say" — but which newspapers? The two most widely used indices, those of Caldara and Iacoviello (2022) and Baker et al. (2016), are built from a fixed panel of English-language, mostly American and British outlets. When a defence-equity price is said to be responding to geopolitical risk, it is responding to a quantity constructed from one particular editorial perspective — the Western one.

The research question of this thesis is: whose perception of geopolitical risk is priced in defence equities? Two candidate answers are possible. Either the information that moves defence-equity prices originates where the war is fought — in Ukrainian and Russian media, which are closer to the battlefield, cover it more intensively, and carry it earlier — and reaches Western prices with the delay that translation and attention impose. Or Western markets price the Western narrative, and local coverage, however earlier or more voluminous, adds nothing once the Western signal is controlled for.

The first challenge is measurement, and it is more severe than it appears. Comparing national media ecosystems requires a corpus that actually contains them. GDELT's Global Knowledge Graph version 1.0, the stream most readily available for this purpose, does not: a representative daily file contains 60,690 records, of which seven carry a `.ru` domain and twenty-one a `.ua` domain. Nationality in that stream can be assigned only from the countries an article *mentions*, which for 88.6% of articles is not the country that published it. Series built on that basis and labelled Ukrainian or Russian sentiment therefore measure the tone of English-language writing *about* Ukraine and Russia — one editorial population sorted by topic and relabelled as three perspectives — and any comparison between them is a comparison of two topic proxies rather than of two vantage points. This thesis uses instead GDELT's *translingual* archive, which processes material in 65 source languages and identifies each article by the outlet that published it. The resulting indices begin on 18 February 2015, the first day of that archive, and cover 4,027 days.

The second challenge is identification. News coverage of a conflict and defence-equity prices may move together for many reasons that have nothing to do with the media ecosystem carrying the signal. Both respond to the same underlying events; financial market conditions affect both; and a common regime trend contaminates any level regression. The strategy addresses this through a conditional joint F-test: local media are asked whether they add to returns *conditional on Western coverage already being in the regression*. This mirrors the test in Bondarenko et al. (2024): they ask whether local Russian risk measures add to English-language measures for the Russian economy; this thesis asks the same question for the counterparty's assets. The market control must also be regional: European defence returns are controlled with the STOXX 600 rather than the S&P 500, a choice that Section 4.4 shows to be decisive rather than cosmetic.

The third challenge is multiple testing. The specification grid crosses frequency (daily, weekly), episode window (six anticipation periods plus the full sample), and five defence-equity targets, producing 31 testable cells. Running 31 tests at 5% significance would yield roughly 1.5 false rejections by chance. Benjamini–Hochberg correction is therefore applied across the full grid. Four tests are reported, referred to throughout as Gates 2 to 5, retaining the numbering of the research design: Gate 2 tests coverage volume and tone, Gate 3 the anticipated/realised structure, Gate 4 European natural gas, and Gate 5 realised conflict escalation. **Gates 3 to 5 are pre-registered; Gate 2 is not.** For Gate 3 the pass rule, the excluded targets and the required sign condition were fixed before the test was constructed, and for Gates 4 and 5 the data were collected only after the pre-registration was committed, so in those two cases the ordering of specification and observation is verifiable from the version history rather than asserted. Gate 2 is reported as the exploratory test it is, and the results below turn on a timing contrast within it rather than on any single cell clearing a threshold.

The fourth challenge is evaluating a null result. A finding that nothing predicts is uninformative unless the magnitude of effect the design could have detected is also established. The out-of-sample section of this thesis therefore accompanies its zero Clark–West rejections with a power simulation: using the actual return series and implanting a predictor with a known true effect, the design is shown to detect an out-of-sample R^2 of 0.5% in 83% of simulated samples and one of 1.0% in every simulated sample, on 1,855 out-of-sample days. The range of effect sizes the predictability literature treats as economically meaningful is therefore ruled out rather than merely unobserved.

The principal finding is that local-language perception adds nothing to defence-equity returns once Western coverage is in the regression, and that the design is nonetheless able to detect media effects where they exist. The second half of that statement is what makes the first a finding rather than an absence, and it is established test by test rather than in aggregate. The Western block survives multiple-testing correction in the pre-registered anticipation test (2 of 31 cells, minimum $p = 1.6 \times 10^{-4}$), and is detected at $p = 4 \times 10^{-5}$ in the natural-gas test. In the attention-and-tone test neither block survives correction under the tradeable one-day lag; sensitivity there is demonstrated instead by the local block's own behaviour, which is detected in seven of 31 cells on the day of publication and in one when the news is lagged by the single day that would have made it tradeable.

The most striking descriptive finding is an asymmetry in how the two sides' media responded to the invasion itself. Russian state media's conflict tone did not move when Russia invaded Ukraine — a shift of +0.02 in aggregate, and −0.05 on a fixed panel of the 24 state outlets present on both sides of the event — while Ukrainian media's tone fell by 1.66 points on the same scale. This is the largest and most robust regularity in the corpus, and it is consistent with the pricing result rather than separate from it: the information set that changed most sharply at the defining event of the sample is not the one whose movements defence-equity prices track.

The null extends from contemporaneous association to forecasting. Across 50 out-of-sample specifications the Clark–West test rejects in zero cases against 2.5 expected by chance, the best out-of-sample R^2 is +0.10%, and no specification reaches even nominal significance. The accompanying power simulation establishes that this reflects the data rather than the design.

The primary contribution is to the literature on whose information is priced in financial markets. Bondarenko et al. (2024) show that for the Russian domestic economy, local Russian-language risk perception matters and English-language perception does not. This thesis runs the mirror of that test on the counterparty's assets and finds the reverse asymmetry: Western defence equities price the Western narrative, and the local narrative is redundant once the Western one is controlled for. Neither result licenses a general claim that local media always matter or never matter. Read together they say something more specific and more useful — that each market prices the information ecosystem its own marginal investors inhabit — and that claim requires both halves of the mirror to state.

The second contribution is to the literature that measures risk from newspaper text, where the question of *which* newspapers has been left largely unexamined. Since Baker et al. (2016) and Caldara and Iacoviello (2022) established normalised newspaper counts as a workable proxy for uncertainty and geopolitical risk, the resulting indices have been treated as measurements of a latent state of the world rather than as measurements taken from a particular vantage point. This thesis makes the vantage point itself the object of study, and shows that the choice is empirically consequential in one direction and not in the other: the Western vantage point carries the pricing information, and adding three further vantage points to it adds nothing that survives correction.

The third contribution is methodological, and concerns how easily this class of measurement goes wrong. Loughran and McDonald (2011) caution that automated sentiment dictionaries require domain-specific validation, which is why GDELT tone is interpreted here as an indicator of coverage negativity rather than as a direct measure of investor beliefs. The same caution applies one level higher, to the corpus rather than the dictionary. A national sentiment series built from an English-only stream measures topic while appearing to measure provenance, and the resulting error is invisible in the output: the series are well behaved, they correlate with events, and they differ from one another. Section 4.4 reports six further specifications that were significant under defensible initial choices and did not survive stricter ones. Both are reported rather than suppressed, because in a literature whose measurements are constructed rather than observed, the conditions under which a result dissolves are as informative as the result itself.

The fourth contribution is to the empirical literature on defence equities and conflict, which is more equivocal than it is often summarised as being. Event studies find sharp price responses at conflict onsets: Covachev and Fazakas (2024) document abnormal returns of up to 12% on European defence stocks in the days following the February 2022 invasion, and Klomp (2025) finds U.S. defence firms gaining 10–15% after the October 2023 attacks on Israel. Evidence that *continuously measured* geopolitical risk predicts defence returns is considerably weaker. Apergis et al. (2018) find that geopolitical-risk shocks predict the volatility of roughly half of 24 leading defence companies but do not predict their returns, and Zhang et al. (2023) similarly locate the effect of geopolitical risk in volatility rather than in the conditional mean across 32 markets. More generally, Tetlock (2007) and Manela and Moreira (2017) establish that media content carries information for asset prices at short and long horizons respectively. What none of this work does is separate news by the nationality of the outlet producing it, which is the separation this thesis performs.

The wider implication concerns an assumption that is increasingly built into commercial risk measurement. Multilingual geopolitical-risk indicators are constructed on the premise that sources closer to an event carry information that Anglophone coverage omits. That premise is testable, and the setting studied here is close to the most favourable one available for testing it: an eleven-year sample, the most intensively covered conflict in modern media history, an asset class whose valuation rests on a claim about that conflict, and local media operating at near-saturation coverage of it. The premise does not hold at any horizon at which the information could have been traded. Proximity to an event, in this setting, does not confer proximity to its price.

2 Data and Identification Strategy

2.1 The news corpus and its construction

The analysis rests on a corpus of 4,027 days of conflict-related news drawn from the Translingual Global Knowledge Graph of GDELT (Leetaru and Schrodt, 2013), covering 18 February 2015, the first day of that archive, to 20 May 2026. The corpus is accessed through BigQuery's `gdelt-bq.gdeltv2.gkg_partitioned` table, a day-partitioned store of GDELT's global news annotations running to several terabytes, from which only the columns and days required are scanned. An article enters the sample if it geolocates to Ukraine or Russia according to GDELT's country codes. GDELT's translingual stream processes text in 65 source languages, machine-translates it, and applies a unified annotation pipeline, making it possible to compare Ukrainian, Russian, and Western outlets through a single measurement framework.

Each article is assigned to one of five media ecosystems based on the outlet that published it — not based on the country the article discusses. This distinction is critical: an article in a British newspaper about Russian troop movements is Western, not Russian. The assignment uses four tiers in priority order. First, a hand-curated register assigns high-volume outlets manually, verified by country of registration and editorial ownership. Second, outlets with a nationally registered domain suffix (`.ua`, `.ru`, `.de`, etc.) are assigned by ccTLD. Third, source language is used conditional on GDELT country information. Fourth, unresolved articles fall to a residual "other" category. Country of publication takes priority over article language at every tier: Ukrainian outlets publish heavily in Russian (e.g. `24tv.ua` carries 2,595 Ukrainian-language and 1,865 Russian-language articles), and classifying them by language would file those articles as Russian and manufacture agreement between two ecosystems the design needs to keep apart. A further rule governs the two cases in which ownership and location diverge, and it is stated explicitly because it must point in both directions to be a rule at all: state-funded external broadcasters are classified to the funding state, which places Deutsche Welle and Radio Free Europe/Radio Liberty in the Western block, while exile newsrooms are classified to their country of origin, which keeps *Meduza* in the Russian independent block. Section 4.4 shows that misapplying this rule to a single high-volume outlet is sufficient to generate a significant result where none exists.

The five resulting ecosystems are: **UA** (Ukrainian outlets), **RU_STATE** (Russian state-controlled), **RU_INDEP** (Russian independent), **WEST** (EU, UK, US, Canada, Australia), and **EN_GLOBAL** (English-language global sources). Russian state outlets are identified through ownership registers following the press-freedom classification in Bondarenko et al. (2024). Russian independent outlets include platforms exiled or blocked in

Russia such as *Meduza* and *TV Rain*.

The register is validated against an external source rather than asserted. Each registered domain is resolved against Wikidata and its recorded country of origin compared with the assigned ecosystem: of 84 registered outlets, 66 resolve to a verifiable country, and 63 of those agree, a precision of **0.955**. The Russian state, Western and Ukrainian blocks agree perfectly (1.000 each); all three disagreements fall in the Russian independent block (0.750), and all three are exile newsrooms, which is the one case where country of origin and country of operation genuinely differ. This audit validates that each registered domain belongs to the country it was assigned to. It does not validate GDELT's attribution of a given article *to* a domain, which is taken on trust and would require a reader with Russian and Ukrainian to check.

The classification rule is also varied, to establish that the results are not an artefact of it. Five rules are run through the full test grid: the baseline, a register-only rule that drops the ccTLD and language tiers, a rule with no language tier, a language-first rule, and one that retains news aggregators. The local-block verdict is unchanged under all five — one or two survivors of 31 cells under the primary alignment in every variant. The language-first rule is the informative failure: because the language tier claims Russian-language articles before ownership is consulted, it cannot represent a Russian independent block at all, which is the concrete cost of the country-before-language rule being got wrong.

2.2 Perception indices

Two indices are constructed for each ecosystem per day. The **attention share** is the count of conflict articles in the ecosystem divided by that ecosystem's total article count, expressed as a percentage. Using a share rather than a raw count is standard practice (Baker et al., 2016) because the overall size of the GDELT corpus drifts over time. The **conflict tone** is the mean GDELT tone score for conflict articles in that ecosystem. GDELT tone is a dictionary-based measure ranging from approximately -10 (highly negative) to $+10$ (highly positive).

Both indices are used in daily first differences in all regressions. The reason is empirical: in levels, all series carry a common "war regime" trend that financial controls already capture, and it is the day-to-day *innovation* in perceived risk, not its persistent level, that should carry price-relevant information. The first-difference transformation is also what Bondarenko et al. (2024) apply.

Following Caldara and Iacoviello (2022), each ecosystem's attention share is further decomposed into an **anticipatory component** (articles classified by GDELT's Themes field as covering threats, build-ups, or pre-war events) and a **realised component** (articles covering acts of war, escalation, or attacks). This threat/act split is the object of the anticipation test in Section 4.1.

2.3 Financial targets and controls

The primary financial targets are two official Bloomberg defence-equity indices: the Bloomberg World Large, Mid and Small Aerospace & Defense Total Return Index (**WAERLST**, USD) and the Bloomberg Europe Defense Select Total Return Index (**BSHIELDT**, EUR), both running from 2 January 2020 to 29 June 2026 (1,619 trading days, no missing observations). Three further targets extend coverage backwards and separate the two regions, giving five in total: the iShares U.S. Aerospace & Defense ETF (**ITA**), available for the whole 2015–2026 window and so covering the pre-war baseline the Bloomberg indices miss; and two hand-built equal-weighted baskets of listed defence names, one European (**EU basket**) and one American (**US basket**), which extend the same coverage to constituents rather than index vehicles. The two baskets are constructed from free data and are the targets the Gate 3 pre-registration excluded from its primary arm, precisely because they are hand-built.

The choice of market control is not a matter of convenience, and Section 4.4 shows it to be the specification decision on which the largest results in this setting turn. Controlling *European* defence returns with the S&P 500 is superficially reasonable, since the two markets are highly correlated over long horizons. Over the build-up window, however, their correlation is only 0.409, so a U.S. control leaves a substantial component of ordinary European market variation in the residual — and that residual correlates 0.26 with the threat measure being tested. Every regression reported here therefore uses the regional market return appropriate to its target: the STOXX 600 for European targets and the S&P 500 for global and U.S. ones. The remaining controls are the VIX, which absorbs general risk appetite, and lagged defence-equity returns, which absorb own-momentum.

2.4 Physical attack data (supplementary context)

The analysis is grounded in media perception rather than physical events, but a parallel dataset records the physical dimension of the conflict: daily Ukrainian Air Force records of Russian air attacks from 29 September 2022 to 21 June 2026. The raw file contains 3,812 attack-level records, aggregated to 809 unique market-information dates using the rule

$$\text{market\_info\_date} = \max(\text{attack\_date},\ \text{time\_end\_date}), \tag{1}$$

which assigns each attack to the day on which the verified count could plausibly enter the market's information set, avoiding look-ahead bias. Across all records, 102,396 weapons were launched and 76,126 were intercepted or destroyed (interception rate: 74.3%); UAVs account for 94.8% of all launched weapons. War intensity is measured as $\ln(1 + \text{weapons launched}_t)$.

This dataset is used in two ways. First, it provides physical context for the media patterns in Section 3: Western news attention spikes on high-intensity attack days, while local attention is already near saturation. Second, it serves as a robustness control in Section 4.4: conditioning the main specification on contemporaneous attack intensity leaves the local-block null unchanged.

The dataset binds the joint sample to September 2022 onward, which is why the primary analysis uses GDELT-only over the longer 2015–2026 window. The physical and perceptual dimensions of the conflict are thus separated in the sample design rather than conflated.

2.5 Identification strategy

The core test asks whether local media add explanatory power to defence returns *conditional on* the Western signal. The specification for each cell in the grid is:

$$r_{i,t} = \alpha + \beta_{\text{West}}\mathbf{X}_t^{\text{West}} + \beta_{\text{Local}}\mathbf{X}_t^{\text{Local}} + \gamma Z_t + \varepsilon_{i,t}, \tag{2}$$

where $r_{i,t}$ is the return on defence-equity target i, $\mathbf{X}_t^{\text{West}}$ contains attention-share and tone changes for the Western and EN_GLOBAL ecosystems, $\mathbf{X}_t^{\text{Local}}$ contains the same for UA, RU_STATE, and RU_INDEP, and Z_t contains the market control and VIX. Standard errors are HAC(5) throughout.

The object of interest is a **joint F-test on** β_{Local}, with the Western block already in the regression. This conditioning is what separates "local media add something" from "local and Western media correlate with returns for the same reason". The test is run across frequency (daily and weekly), six conflict-episode windows and the full sample, and five targets — producing 31 cells that survive the minimum-observation rule. Benjamini–Hochberg (BH) correction (Benjamini and Hochberg, 1995) is applied across all 31 cells simultaneously.

Two conventions recur in every results table and are defined here once. "BH survivors" is the number of cells that remain significant after Benjamini–Hochberg correction (Benjamini and Hochberg, 1995) controlling the false discovery rate at 5% across all cells of a grid simultaneously; throughout this thesis it is that count, not the number of nominally significant cells, that decides a verdict. "HAC(5)" denotes Newey–West standard errors (Newey and West, 1987) with a five-day bandwidth, which remain valid under the serial correlation and heteroskedasticity that daily return regressions exhibit.

The **positive control** is the identical joint F-test on β_{West} in the same 31 cells. If the design detects the Western block, a null on the local block is informative. If it detects neither, the absence of the local result is merely consistent with low power.

The out-of-sample test in Section 4.3 uses an expanding-window design with a minimum 250-day training window, re-estimated at each step and forecasting one day ahead.

The choice of test statistic for that exercise follows from the structure of the comparison and is not interchangeable. Forecast-accuracy comparisons in this literature are most often reported using the Diebold–Mariano statistic (Diebold and Mariano, 1995), which is not valid here. Each of the 50 specifications compares a model — the historical mean augmented with one perception predictor — against the historical mean alone, and setting that predictor's coefficient to zero recovers the benchmark exactly. The benchmark is therefore nested within the model, and under nesting the Diebold–Mariano statistic has no standard normal limiting distribution and is biased against the larger model even when its additional predictor carries genuine information. Clark and West (2007) correct this by adjusting for the estimation noise that nesting introduces, yielding a statistic that is valid in this setting and one-sided in the direction of improvement. Diebold–Mariano with the Harvey et al. (1997) small-sample correction is retained for genuinely non-nested comparisons, but every cell of the reported grid is nested and is arbitrated by Clark–West.

3 Descriptive Statistics

The descriptive evidence establishes two things before any regression is run: that the local and Western media ecosystems are not measuring the same thing, and that the one series which moved most sharply at the invasion belongs to the ecosystem that turns out not to be priced. Table 1 defines every variable used in the analysis and Table 2 reports its distribution; Table 3 reports how the series co-move; Figures 1–3 plot the outcome and the regressors over the full sample; and Table 4 isolates each ecosystem's tone response to the invasion, the event with the greatest identifying power in the sample.

Two features of Table 2 carry most of the descriptive argument. The first is in Panel B: Ukrainian, Russian state and Russian independent outlets devote 71–85% of their output to this conflict on an average day, against 7.2% for Western outlets and 5.4% for native-English ones. That is an order-of-magnitude difference in editorial attention, sustained across eleven years, and it is the sense in which the local ecosystems are “closer” to the war. The local series are also far less variable around those means — Ukrainian attention has a standard deviation of 5.0 percentage points on a mean of 81.0 — because an outlet already devoting four-fifths of its output to a subject has little room left to escalate.

The second feature pre-empts the most obvious objection to the results that follow. Panel D reports the daily changes that are the actual regressors, and shows that the local series are not inert. Those changes are mean-zero by construction, so it is their dispersion that matters: the daily change in Ukrainian attention has a standard deviation of 2.3 percentage points and that in Russian independent attention 4.1, against 1.2 for Western outlets and 1.0 for native-English ones. The local regressors therefore vary two to four times as much from day to day as the Western ones. Whatever explains the null in Section 4, it is not an absence of variation in the local regressors.

Figure 1 plots the outcome variable, and shows that the war repriced defence equities in two distinct steps rather than one. The first step is the invasion itself: in the five weeks from 24 February 2022 the European index gains 21.6%, against 7.0% globally and 9.1% in the United States. The second, and much larger, step comes three years later, when European governments announced rearmament programmes: the European index gains 63.7% in the first half of 2025 alone and ends the sample at roughly 2.5 times its January 2020 level, while the global and U.S. indices end near 1.9 times. Two features of this figure matter for what follows. The European index is the one that responds to the conflict, which is why the regional market control in Section 2.5 is a European rather than a U.S. index. And the largest single move in the sample is driven by European fiscal announcements rather than by battlefield news, which is a reason to test war-news predictability against a market control rather than in isolation.

Table 1: Variable definitions

Variable	Unit	Definition
Financial outcomes		
WAERLST return	%	Daily log return on Bloomberg World A&D Total Return Index.
BSHIELDT return	%	Daily log return on Bloomberg Europe Defense Select Total Return Index.
ITA return	%	Daily log return on iShares U.S. Aerospace & Defense ETF.
Perception indices (one per ecosystem per day; full 2015–2026 sample)		
Attention share	%	Conflict articles as a share of the ecosystem's total daily output.
Conflict tone	score	Mean GDELT tone of conflict articles (range ≈ -10 to $+10$; more negative = more negative coverage).
Anticipatory share	%	Conflict articles classified by GDELT Themes as covering threats or build-ups, as a share of that ecosystem's conflict articles. The "threat" half of the Gate 3 split.
Realised share	%	Conflict articles classified as covering acts of war, attacks or escalation, same denominator. The "act" half of the Gate 3 split.
Non-market outcome		
GPR_ACT	index	The realised-conflict component of the Caldara and Iacoviello (2022) geopolitical risk index; the outcome in Gate 5, where no asset price is involved.
Physical attack variables (robustness subsample only, Sep 2022 – Jun 2026)		
Attack intensity	log count	$\ln(1 + \text{weapons launched}_t)$ on the market-information date; zero on non-attack days.
Interception rate	ratio	Weapons destroyed divided by weapons launched; captures attack effectiveness.
Financial controls		
Market return	%	STOXX 600 (European targets) or S&P 500 (global/U.S. targets) daily log return.
VIX	index pts	CBOE Volatility Index (levels).
TTF gas return	%	Daily log return on Dutch TTF month-ahead natural gas; the Gate 4 outcome.

Notes: All perception indices are taken in daily first differences in regressions, for the reason given in Section 2. Ecosystems are UA (Ukrainian), RU_STATE (Russian state-controlled), RU_INDEP (Russian independent), WEST (EU, UK, North America, Australia), EN_GLOBAL (English-language global). See Section 2 for classification details.

Table 2: Descriptive statistics

Variable	Mean	Std. dev.	Min	Max	N
Panel A: financial outcomes and controls, daily % returns					
WAERLST return	0.05	1.55	−13.74	10.56	1,619
BSHIELDT return	0.07	1.79	−13.41	12.14	1,619
ITA return	0.05	1.42	−15.95	11.60	2,837
STOXX 600 return	0.02	1.03	−12.19	8.07	2,782
VIX (index points)	18.38	7.05	9.14	82.69	2,837
Panel B: conflict attention share (% of the ecosystem's daily output)					
Ukrainian	81.0	5.0	60.1	97.1	3,073
Russian state	70.6	9.3	45.4	92.7	3,073
Russian independent	84.8	7.0	41.3	99.7	3,073
Western	7.2	4.5	2.7	47.2	3,073
Native-English	5.4	2.5	2.3	32.2	3,073
Panel C: conflict tone (GDELT score, ≈ −10 to +10)					
Ukrainian	−2.39	0.63	−5.05	0.78	3,073
Russian state	−1.78	0.45	−3.60	2.03	3,073
Russian independent	−2.31	0.65	−5.50	0.54	3,073
Western	−1.80	0.51	−4.01	1.32	3,073
Native-English	−1.99	0.49	−4.20	−0.38	3,073
Panel D: the actual regressors — daily changes in the Panel B and C series					
Δ Ukrainian attention (pp)	0.00	2.3	−12.4	10.8	3,072
Δ Russian state attention (pp)	0.00	3.2	−18.6	21.3	3,072
Δ Russian indep. attention (pp)	0.00	4.1	−28.3	35.5	3,072
Δ Western attention (pp)	0.00	1.2	−10.4	21.9	3,072
Δ Native-English attention (pp)	0.00	1.0	−7.0	12.0	3,072
Δ Ukrainian tone (points)	0.00	0.42	−2.08	1.84	3,072
Δ Russian state tone (points)	0.00	0.37	−3.31	2.66	3,072
Δ Western tone (points)	0.00	0.40	−1.93	1.46	3,072
Panel E: physical attacks, attack days only (robustness subsample)					
Weapons launched	126.6	144.2	1	982	809
Interception rate	0.735	0.211	0.000	1.000	809

Notes: Produced by `scripts/make_thesis_descriptives.py`. Each series is summarised over the full span on which it exists: the two Bloomberg indices from 2 January 2020, the news series from 18 February 2015 to 20 May 2026. Panel E covers the 809 market-information dates carrying at least one recorded weapon launch, between 29 September 2022 and 21 June 2026. Panel D reports the daily changes of the Panel B and C series, which are the variables that actually enter equation (2); attention changes are in percentage points and tone changes in index points. Two of the ten change series (Russian independent and native-English tone) are omitted from Panel D for space and are reported in the accompanying CSV.

Table 3: Correlations between media ecosystems and defence-equity returns (daily first differences)

	Between media ecosystems					With defence returns	
	UA	RU_ST	RU_IN	WEST	EN_GL	WAERLST	BSHIELDT
Panel A: change in conflict attention share							
Ukrainian (UA)	1.00					−0.01	0.01
Russian state (RU_ST)	0.47	1.00				0.02	0.02
Russian independent (RU_IN)	0.05	0.13	1.00			−0.03	−0.02
Western (WEST)	0.13	0.16	0.15	1.00		−0.03	−0.03
Native-English (EN_GL)	0.02	0.06	0.09	**0.60**	1.00	−0.00	−0.04
Panel B: change in conflict tone							
Ukrainian (UA)	1.00					−0.00	−0.02
Russian state (RU_ST)	0.39	1.00				0.03	0.05
Russian independent (RU_IN)	0.35	**0.57**	1.00			0.03	0.04
Western (WEST)	0.36	0.35	0.33	1.00		0.02	−0.01
Native-English (EN_GL)	0.26	0.28	0.28	**0.58**	1.00	0.02	0.01

Notes: Pearson correlations of daily first differences, the transform used in every regression in this thesis. Correlations are computed pairwise: the media-versus-media block uses all 3,072 corpus days with a lagged value available, while the two right-hand columns use the 936 days on which the Bloomberg indices also trade. WAERLST is the Bloomberg World Aerospace & Defense index, BSHIELDT the Bloomberg Europe Defense Select index. Only the lower triangle is shown; the matrix is symmetric.

Two patterns matter for what follows. **First, the ecosystems are distinct populations rather than one signal measured five times.** The largest off-diagonal correlation anywhere in the table is 0.60, between Western and native-English attention — two ecosystems that overlap by construction, since many global English-language outlets are also Western. Ukrainian and native-English attention correlate at 0.02. Under the previous, topic-based classification these series were near-duplicates; they are not near-duplicates now, which is what makes a conditional test between them meaningful.

Second, no ecosystem is strongly correlated with defence returns unconditionally. Every entry in the two right-hand columns lies between −0.04 and 0.05. This is a necessary but not sufficient condition for the null reported in Section 4: an unconditional correlation of this size could still mask a conditional effect, which is why the tests that follow condition on the Western block and the market control rather than reading the answer off this table.

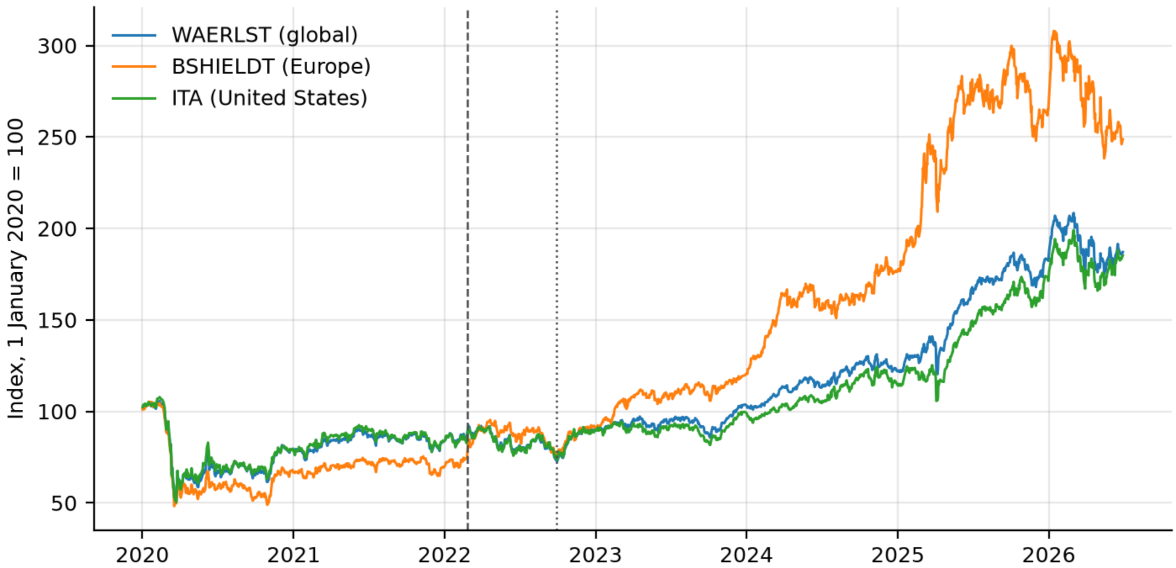

Figure 1: Defence-equity index levels, 1 January 2020 = 100, to June 2026. **WAERLST** (blue) is the Bloomberg World Aerospace & Defense Total Return Index; **BSHIELDT** (orange) is the Bloomberg Europe Defense Select Total Return Index; **ITA** (green) is the iShares U.S. Aerospace & Defense ETF. The dashed vertical line marks the full-scale invasion (24 February 2022); the dotted vertical line marks 29 September 2022, the first day of the Ukrainian Air Force attack record used in Section 2.4. All three series are total-return indices, so the vertical distance between them is cumulative performance including dividends, not a price-only comparison.

Figures 2 and 3 plot the perception indices that are the regressors of interest. The most striking feature of Figure 2 is that the two media environments occupy entirely different levels throughout the eleven years. Ukrainian and Russian outlets devote 71–85% of their output to conflict-related content on an average day and never fall below 41%, while Western and native-English outlets average 7.2% and 5.4% and stay in single digits until the invasion. This is not two media environments covering the same war at different intensities — it is two media environments covering different worlds. The invasion is visible as a step change in the Western lines, from 19.0% on 23 February 2022 to 40.9% on the 24th, and in the native-English lines, from 11.8% to 23.8%. In the same two days Ukrainian attention moves from 90.3% to 95.8% and Russian state attention from 84.6% to 91.2%: the local ecosystems had almost no room left to react, which is the mechanical reason a Western investor watching coverage volume would have seen a signal in Western outlets and nothing much in local ones.

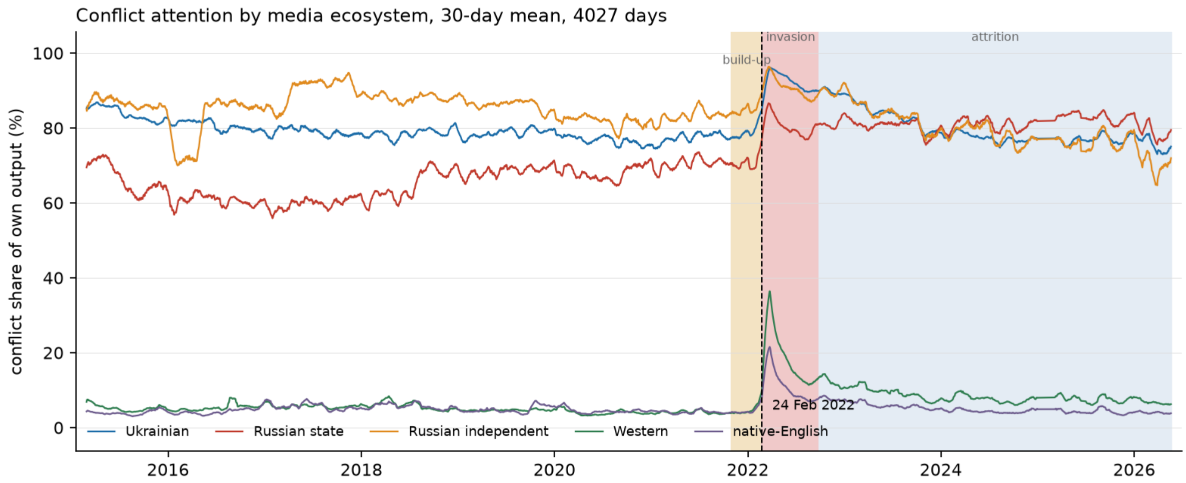

Figure 2: Conflict attention share by media ecosystem, 30-day moving average, 18 February 2015 – 20 May 2026. Each line shows the share of each ecosystem's daily article output that covers conflict in Ukraine or Russia. Shading marks four war regimes: pre-war (white), build-up (light grey, Nov 2021 – Feb 2022), invasion (dark grey, Feb – Jun 2022), and attrition (medium grey, Jul 2022 onward). The dashed vertical line marks 24 February 2022.

Figure 3 reveals a striking asymmetry at the invasion. **Ukrainian conflict tone falls sharply and stays down; Russian state tone barely moves.** Before 2022 the five tone series interleave with no stable ordering; after 24 February 2022 the Ukrainian line separates downward (by -1.66 points; see Table 4) and remains separated for the rest of the sample, while the Russian state line stays within its pre-war band.

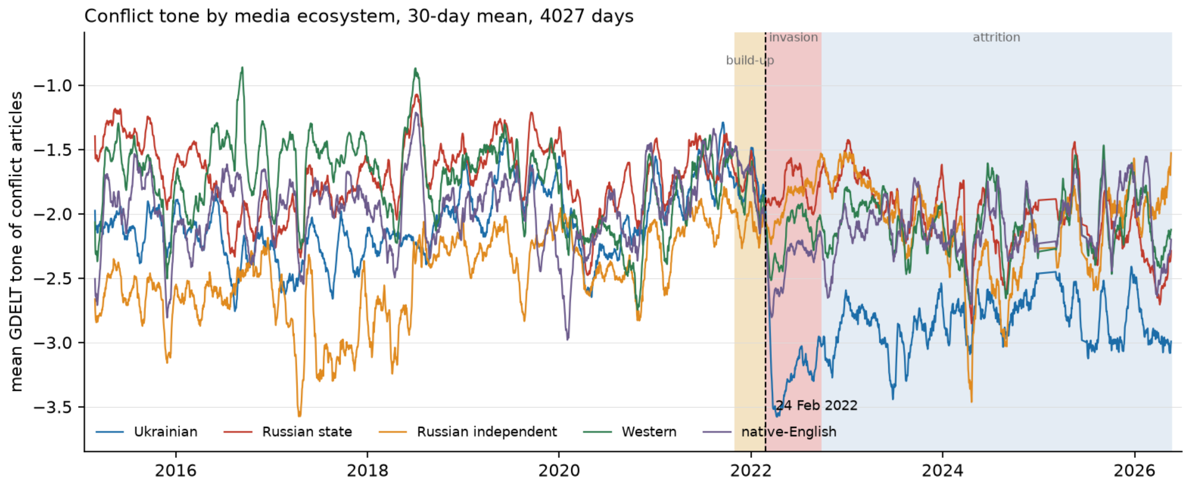

Figure 3: Mean conflict tone by media ecosystem, 30-day moving average, same sample and regime shading as Figure 2. GDELT tone ranges from -10 (most negative) to $+10$ (most positive). Before 2022 the five series interleave with no stable ordering. After 24 February 2022 the Ukrainian line separates downward and stays separated; the Russian state line does not move out of its pre-war band.

Russian state media's tone did not change when Russia invaded Ukraine, and Table 4 quantifies how far that is from what every other ecosystem did. Ukrainian tone falls by 1.66 points, native-English by 0.64, Western by 0.36; Russian state tone moves by +0.02, which is to say not at all. The result is not an artefact of which outlets happen to be in the corpus on either side of the invasion: on a fixed panel of the 24 state outlets present in both windows with at least 200 conflict articles in each, the mean shift is −0.05. Several state outlets in fact became *more positive* — *Gazeta.ru* (+0.48), *Ren.tv* (+0.43), *mskagency.ru* (+0.40) — which is what one would expect from outlets covering the launch of an operation their government had announced as a success. This contrast is the largest and most robust descriptive fact in the data, and it is the reason the pricing null that follows is interesting rather than merely negative: the ecosystem whose signal changed most sharply at the defining event of the sample is not the ecosystem whose signal the market tracks.

Table 4: Conflict tone before and after the full-scale invasion, by ecosystem

Ecosystem	Pre-invasion (Nov 2021–23 Feb 22)	Post-invasion (24 Feb–5 Jun 22)	Shift
Ukrainian (UA)	−1.77	−3.43	**−1.66**
Native-English (EN_GLOBAL)	−1.87	−2.51	−0.64
Western (WEST)	−1.90	−2.27	−0.36
Russian state (RU_STATE)	**−1.81**	**−1.79**	**+0.02**
Russian independent (RU_INDEP)	−2.09	−2.01	+0.09

Notes: Mean GDELT tone score over the two windows named in the column headings (115 pre-invasion and 102 post-invasion days); more negative values indicate more negative coverage. Ecosystems are ordered by the size of the shift. Produced by `scripts/make_thesis_descriptives.py`.

The comparison this table is used for is Ukrainian against Russian state, and that contrast is large and robust. Ukrainian tone falls by 1.66 points; Russian state tone does not move. The Russian state figure is not an artefact of which outlets happen to be in the corpus on each side of the invasion: on a fixed panel of the 24 state outlets present in both windows with at least 200 conflict articles in each, the mean shift is −0.05, the same magnitude as the aggregate. A supremum-Wald test for a break at an unknown date, free to choose any day in the interior 70% of the sample, puts Ukrainian tone's largest break on exactly 24 February 2022, while Russian state tone's largest break falls 393 days later.

The comparison this table is *not* used for is Russian state against Russian independent. The two differ by 0.07 points in the aggregate and by 0.11 on the fixed panel, and a formal test of that difference does not reject ($p = 0.561$). A gap of this kind is readily interpreted as a censorship wedge, and Section 4.4 reports why that interpretation is not sustainable here. The Russian independent block is small — four outlets clear the fixed-panel threshold against 24 state outlets — which is the proximate reason the comparison lacks power.

4 Empirical Findings, Interpretation, and Implications

4.1 Gates 2 and 3: local perception and defence-equity returns

Local media add nothing to defence-equity returns that Western media do not already carry, in either of the two forms the question can be asked. Table 5 reports both. Gate 2 asks it of raw coverage volume and tone — does the local ecosystem's attention or sentiment move returns? Gate 3 asks it of structure — does the local split between anticipated and realised conflict, which media at war record very differently from media watching one, move returns? Gate 2 is exploratory. Gate 3 is pre-registered, with its pass rule, its excluded targets and its sign condition fixed in a committed document before the test was estimated — a distinction that matters here, because Gate 3 initially cleared that rule on the data available at the time and failed it once the held-out window was added.

Table 5: Local-media joint F-tests on defence-equity returns (Gates 2 and 3)

Gate	Block and news timing	Cells	BH survivors	Min. p	Verdict
Gate 2: coverage volume and tone					
2	Local, news lagged 1 day *(primary)*	31	1	9.0×10^{-4}	FAIL
2	Local, same-day	31	**7**	$\mathbf{2.2 \times 10^{-6}}$	—[a]
2	*Western, news lagged 1 day (control)*	31	0	2.7×10^{-3}	—
2	*Western, same-day (control)*	31	0	1.8×10^{-2}	—
Gate 3: anticipated/realised split (pre-registered)					
3	Local, news lagged 1 day *(primary arm)*	31	2	5.5×10^{-6}	FAIL[b]
3	Local, same-day *(secondary arm)*	31	0	—	FAIL
3	*Western, news lagged 1 day (control)*	31	**2**	$\mathbf{1.6 \times 10^{-4}}$	PASS

Notes: Each cell is a joint F-test on one block of coefficients in equation (2), with the other block, the regional market control and VIX already in the regression; HAC(5) standard errors. The grid crosses daily/weekly frequency, six conflict-episode windows plus the full sample, and five defence-equity targets, giving 31 cells that clear the minimum-observation rule. Benjamini–Hochberg correction at 5% FDR is applied across all 31 cells of an arm simultaneously. "Min. p" is the smallest uncorrected p-value in the arm, reported so that the survivor count can be read against the strength of the strongest single cell. **Lagged is the primary timing convention**: GDELT days are full UTC days while European markets close at 16:30 UTC, so only news credited to the following trading day was reliably available to trade on.

[a] **This row is the informative one, and it is not evidence for local pricing.** Seven of 31 same-day cells survive correction, at a minimum p of 2.2×10^{-6} — so the design is emphatically able to detect the local block when the local block is there to detect. Crediting the same news to the next trading day, which removes only the portion published after the European close, reduces those seven survivors to one. Information the market had not yet impounded should have survived that shift; what does not survive it is the mechanical same-day association that efficient pricing predicts.

[b] The two Gate 3 primary survivors are both weekly and both in the 2025–26 episode window (58 observations against 13 parameters). The two deepest cells in the grid, the pooled daily specifications with 2,754 observations each, return $p = 0.08$ and $p = 0.36$. Neither pre-registered arm clears: Arm 1 requires a surviving cell in the Russia build-up window on a Bloomberg target or ITA, and on the corrected register that window produces no surviving cell at all (smallest p there is 0.018); Arm 2 requires same-sign survivors in two independent episode windows, and the two that survive share a window.

The timing contrast in Gate 2 is the result that rules out low power as an explanation of the null. Local coverage is strongly associated with defence returns when credited to its publication day: seven of 31 cells survive correction, the strongest at $p = 2.2 \times 10^{-6}$. Credited instead to the following trading day — the only convention under which the information could have been acted upon, since GDELT days run to midnight UTC while European markets close at 16:30 — seven survivors become one. The lag displaces only the portion of coverage published after the close. Information not yet impounded in prices should therefore have survived it largely intact, whereas a mechanical contemporaneous association should not. The observed pattern is the one efficient pricing implies, and estimating a single alignment would have left the two indistinguishable.

Gate 3 reaches the same verdict on the structural version of the question, and its two survivors fail the pre-registered rule for an instructive reason. They are both weekly cells in the 2025–26 episode window, fitting 13 parameters to 58 observations, while the two deepest cells in the grid — pooled daily specifications with 2,754 observations each — return $p = 0.08$ and $p = 0.36$. An effect that concentrates in the thinnest corner of a grid and disappears where the data are deepest is the signature of noise rather than of a small true effect awaiting more power. The Western block, tested in the same 31 cells under the same correction, does survive, at a minimum p of 1.6×10^{-4}. That asymmetry — Western detected, local not, in identical cells — is the finding.

Gate 3 also produced a coefficient pattern with an economic interpretation, which a held-out test did not support. Among the surviving weekly cells the Ukrainian coefficients were consistent with "buy the rumour, sell the fact": a shift toward anticipatory coverage raised defence returns and a shift toward realised conflict lowered them. Those four signs were recorded and then tested on the 2017–19 window, which had not been examined when the hypothesis was formed. Seven of twelve signs matched, against six expected under the null (binomial $p = 0.387$). The pattern was in-sample noise. The episode illustrates a general point about pre-registration in this setting: a test fixed in advance but estimated on a sample that is still being extended is not, in the relevant sense, pre-registered, because the data that would discipline it have not yet been observed.

4.2 Gates 4 and 5: gas and escalation

Gates 4 and 5 move the test to the two outcomes where local information should matter most, and the null survives both. If the defence-equity result were an artefact of defence equities being a poor testbed — which is a fair objection, since the link from a Russian newspaper to a U.S. contractor's share price runs entirely through Western investors — then it should break on an asset with a direct physical transmission channel, or on an outcome with no market in it at all. Table 6 tests both. European natural gas is the asset through which the conflict physically transmitted to Europe (Russia supplied

~40% of EU gas; TTF rose from ~€20 to ~€300); realised escalation is the war outcome itself. Both tests were pre-registered before their data were ingested.

Table 6: Local-media joint tests on European gas (Gate 4) and realised escalation (Gate 5)

Gate / Window	n	p (local block)	p (Western block)
Gate 4: Dutch TTF natural gas, daily returns			
(a) Build-up and invasion, Jun 2021 – Jun 2022	222	0.399	**0.00004**
(b) Shutdown and aftermath, Jun 2022 – Jun 2023	231	0.533	0.193
(c) Full crisis, Jun 2021 – Dec 2023	563	0.250	0.066
Gate 5: realised escalation (GPR_ACT), held-out sample			
Horizon $h = 1$ day	637	0.212	0.171
Horizon $h = 5$ days	629	0.227	**0.033**

Notes: Each entry is the p-value of a joint F-test on one block — local (Ukrainian, Russian state, Russian independent attention and tone changes) or Western — with the other block and the controls already in the regression; HAC(5) standard errors. Bold marks $p < 0.05$. Gate 4 regresses daily TTF returns; Gate 5 regresses the realised-conflict component of the Caldara and Iacoviello (2022) index on 637 held-out days that were never used in sample. Both tests were pre-registered, and in both cases the data were collected after the pre-registration was committed, so the ordering is verifiable from the version history.

All four of Gate 4's pre-registered conditions fail, and they fail in different ways, which is what makes the verdict robust rather than marginal: no BH survivor in either required window; the placebo condition breached, since local perception "explains" Brent crude ($p = 0.0014$) as readily as European gas; the build-up result does not survive dropping the ten largest TTF moves ($0.399 \to 0.840$); and the ordering is wrong, with Russian *independent* rather than Russian *state* media leading the local block, which inverts the supply-signalling mechanism the test was built to detect.

The Western block behaves differently in the two gates, and the honest reading is that Gate 4 carries the positive control and Gate 5 carries it only weakly. In Gate 4's build-up window the Western block is detected at $p = 4 \times 10^{-5}$ in the same specification where the local block returns 0.399 — the sharpest positive control anywhere in this thesis. In Gate 5 the Western block is significant at the five-day horizon ($p = 0.033$) but not at one day ($p = 0.171$), so Gate 5 on its own would not establish that the design has power; it is reported as corroboration, not as an independent control. A shuffle placebo returns $p = 0.557$ and 0.799, confirming the procedure does not manufacture significance from noise.

Gate 4 shows that the null is not a property of defence equities being a weak testbed. European gas is the asset where Russian supply signals should matter most — Russia supplied roughly 40% of EU gas and TTF moved from about €20 to over €300

— and Russian state media is the channel through which supply intent was signalled. Yet the local block returns $p = 0.399$ in the build-up window, and all four pre-registered conditions fail. In the same specification the Western block is detected at $p = 4 \times 10^{-5}$, which is the clearest demonstration in this thesis that the apparatus finds effects where they exist.

Gate 4 also demonstrates how a significant result in this setting can dissolve under replication alone. Estimated exploratorily on the 81 days of gas-crisis coverage available at the time, the build-up specification returned $p = 0.0028$ under the controls the confirmatory test would later use, and $p = 0.002$ under an alternative control set — stable enough across specifications to appear robust. The pre-registered confirmatory estimate, on the same asset, specification, controls and calendar period, with nothing altered but the quantity of data, returned $p = 0.399$ on 222 continuous days. The result was not overturned by a better control or a stricter correction, but by fuller coverage of the period it had always concerned. This is the failure mode that pre-registration is designed to expose, and it is invisible to any robustness check conducted within the smaller sample.

Gate 5 removes prices from the test entirely and asks whether local perception anticipates the war's own escalation. If local media carry genuine information about the conflict, they should at minimum lead a measure of realised conflict, whether or not any asset prices it. On 637 held-out days never used in sample they do not: the local block returns $p = 0.21$ at one day and $p = 0.23$ at five, and does not survive twelve lags of the outcome's own dynamics. This gate is weaker evidence than the other three, because its own Western control is significant at five days ($p = 0.033$) but not at one ($p = 0.171$); it is reported because a pre-registered test must be reported whatever it returns, and because a null on a non-market outcome closes the escape route that the earlier nulls are all artefacts of market microstructure.

4.3 Out-of-sample predictability

None of the preceding tests is a statement about forecastability, which requires separate evidence. A contemporaneous association may exist without being exploitable, and equally an absence of contemporaneous association does not establish that no predictive relationship exists at any horizon. Table 7 reports the out-of-sample results together with the power simulation that makes their null interpretable. The forecasting grid has five targets and ten predictors (attention share and tone for each of five ecosystems), giving 50 specifications. The Clark–West test (Clark and West, 2007) arbitrates each. Where models are non-nested, the Diebold–Mariano test (Diebold and Mariano, 1995) with the Harvey–Leybourne–Newbold small-sample correction (Harvey et al., 1997) is used instead.

Table 7: Out-of-sample return predictability and test power

Panel A: Clark–West results across the full grid		
Specifications (5 targets × 10 predictors)		50
Out-of-sample days per specification		1,855 or 686
Positive Campbell–Thompson R^2_{OS}		3 of 50
Best R^2_{OS} observed	+0.0010	(+0.10%)
Smallest Clark–West p-value		0.067
Clark–West $p < 0.05$	**0**	(2.5 expected by chance)
Survive Benjamini–Hochberg (5% FDR)		**0**
Panel B: Simulated power, 1,855 OOS days, 60 paths per row		
True R^2_{OS} = 0.0% (nominal size)		0.02
True R^2_{OS} = 0.2%		0.47
True R^2_{OS} = **0.5%**		**0.83**
True R^2_{OS} = 1.0%		1.00
True R^2_{OS} = 2.0%		1.00

Notes: Expanding-window, one-day-ahead design with a minimum 250-day training window, re-estimated at each step. Each specification adds one perception predictor — the lagged daily change in attention share or tone for one ecosystem — to a historical-mean benchmark. Because setting that predictor's coefficient to zero returns the benchmark exactly, the comparison is nested and is arbitrated by the Clark–West test (Clark and West, 2007) with a Bartlett long-run variance estimator, one-sided in the direction of model improvement. Out-of-sample length differs by target: the three free-data targets yield 1,855 out-of-sample days each and the two Bloomberg indices, which begin in 2020, yield 686.

Panel B simulates power by implanting a predictor with a known true R^2_{OS} into the actual return series and running the identical expanding-window machinery over 1,855 out-of-sample days. **The simulation uses 60 paths per row, so each rejection rate carries a standard error of roughly six percentage points**; the entries should be read as locating the power curve, not as precise to the second decimal. Nominal size in the first row is 0.02 against a nominal 5%, confirming the test is conservative rather than over-rejecting — which means the null in Panel A is not produced by a test that cannot reject.

Zero Clark–West rejections across 50 specifications is a bound, not an absence.

Panel B quantifies what the null rules out. The 0.5–1.0% range of out-of-sample R^2 that Welch and Goyal (2008) and Campbell and Thompson (2008) treat as the threshold for economic significance is detected in 83% of simulated samples at the bottom of that range and in all of them at the top. The best value actually observed, +0.10%, sits below even the 0.2% effect size that the design detects only about half the time — and its Clark–West p-value is 0.067, so not one of the fifty specifications reaches nominal significance, let alone survives correction. What can be claimed is therefore bounded in both directions: an effect large enough to matter economically would very probably have been seen, and an effect small enough to have escaped this design is too small to trade.

4.4 Discussion: what this null means

The null is consistent across all four tests and both asset classes. Local perception adds nothing in coverage volume, in tone, in the anticipated/realised structure, in the asset with the most direct transmission channel, in a non-market outcome, or out of sample. The design is nonetheless demonstrably sensitive: the Western block is detected under multiple-testing correction in the anticipation test and at $p = 4 \times 10^{-5}$ in the gas test, and in the attention-and-tone test the local block itself is detected in seven of 31 cells before the tradeable lag is imposed. The informative feature of these results is their co-occurrence — a null on the local block in precisely the cells where an effect is detected on the Western block, or on the local block at an untradeable alignment. Media effects on prices are measurable in this design; the local-language ones are not among them.

Western investors respond to the signal that moved, and the descriptive chapter shows which signal that was. Russian state media's tone did not change when Russia invaded Ukraine, and Ukrainian and Russian attention were already near saturation and had almost no room to rise. Western attention, by contrast, doubled in a day. If the marginal buyers and sellers of Western defence equities form their views from Western coverage — which is what the null says — then they responded to the one series in the data that registered the event. The information asymmetry between the two media environments is real and large; what does not follow is that the asymmetry advantages local media from the standpoint of a Western investor. Coverage that is already saturated cannot rise when something happens, and a signal that does not move cannot move a price.

The result runs against the prior that motivated the design, which is stated here explicitly. Local media are closer to the events, cover them earlier and in far greater volume, and Bondarenko et al. (2024) had already established that local-language risk measures carry information English-language measures do not. The prior was therefore that local perception would contribute something, most plausibly through the anticipated/realised structure tested in Gate 3, and the pre-registrations were specified to detect such a contribution rather than to rule one out. Gates 4 and 5 were added because defence equities are a weak testbed for this question, the transmission from Russian-language reporting to a Western contractor's share price running entirely through Western investors. Neither the primary tests nor the two additional ones detect a contribution.

Reconciling that with the literature requires being precise about what Bondarenko et al. (2024) actually established. Their finding is that local-language geopolitical-risk shocks move the *Russian* economy while English-language shocks do not. The natural extrapolation — local sources are simply better measures of geopolitical risk — is the one this thesis rejects. What survives is narrower and more useful: the informative ecosystem is the one inhabited by the agents whose decisions are being measured. Russian firms and households read Russian newspapers, so Russian coverage moves Russian output; Western fund managers read Western newspapers, so Western coverage moves Western defence equities. Read that way the two results are not in tension at all, and neither is a claim about which journalists are better informed.

The results refine rather than contradict the defence-equity literature, and they sit more comfortably within it than a null might suggest. Covachev and Fazakas (2024) and Klomp (2025) document large abnormal returns on defence stocks at conflict onsets, and nothing reported here disputes that: the Western block is detected in the anticipation and gas tests, and Figure 1 shows the invasion repricing plainly. What this thesis adds is that the response is specific to the Western reporting layer rather than to conflict news as such — a distinction invisible to any design that aggregates coverage without regard to the publisher, as all of these studies do, and invisible for a practical reason, since separating publishers requires a translingual corpus available only for part of the period they cover.

The finding that continuous news measures do not predict returns is, if anything, the established position rather than a departure from it. Apergis et al. (2018) reach the same conclusion for returns on 24 defence firms using the published geopolitical-risk index and a nonparametric design far removed from the one used here. Where their result and this one appear to diverge is on volatility: they find geopolitical risk predicting the volatility of about half their firms, whereas Section 4.4 reports that war variables fail to improve on HAR and GARCH benchmarks. The two are compatible, because the questions differ. Theirs is whether geopolitical risk carries in-sample information about volatility; the one asked here is whether war-specific news adds *out-of-sample* forecasting accuracy to volatility models that already exploit the persistence of realised variance. A variable can satisfy the first and fail the second, and conflict news evidently does.

The physical record of the war does not forecast defence-equity returns either, which closes off the reading that the news measures are simply a poor proxy for the war itself. On the September 2022 – June 2026 subsample where the Ukrainian Air Force attack data exist, 144 specifications race five information sets — physical attacks alone, news alone, and their combinations — against historical-mean and AR(1) benchmarks across three targets, two horizons and two samples. Sixteen achieve a positive out-of-sample R^2 and **none survives Benjamini–Hochberg correction**; the benchmarks outperform in the majority of cells. Realised physical intensity is no more informative about subsequent defence-equity returns than perceived intensity is, which indicates that the null documented above reflects the pricing relationship rather than a deficiency of the news measures as proxies for the conflict.

War signals do not forecast defence-equity volatility either, which is the specification in which a positive result was most plausible. Returns are close to unforecastable in efficient markets, whereas volatility is persistent and routinely forecastable, so conflict information that carries no signal for the conditional mean might still carry one for the conditional variance. Forty-eight augmented HAR-RV-X specifications (Corsi, 2009) are evaluated against GARCH, GJR-GARCH and EGARCH benchmarks under the QLIKE loss. **None improves significantly on its benchmark and eight are significantly worse**, with all six specifications surviving multiple-testing correction being deteriorations. Augmenting a volatility model with war variables degrades it. Both exercises reported in this subsection are exploratory rather than pre-registered, and are identified as such.

Six specifications in this setting yield significance under defensible initial choices and lose it under stricter ones, and the six mechanisms are distinct. Two fail on an omitted variable: a defence-volatility result and a build-up threat result, each estimated with the S&P 500 as the market control for European defence equities, and each disappearing when the STOXX 600 replaces it, the second moving from $p = 0.0001$ to $p = 0.84$. One fails on sample truncation: the Gate 3 grid clears its pre-registered rule on a partial sample and fails once the held-out window is included, seven surviving cells becoming two. One fails on replication with more data: the gas specification, $p = 0.0028$ on 81 days and $p = 0.399$ on 222 days of the same calendar period. One fails out of sample: an escalation result significant in both halves of the in-sample period and $p = 0.21$ on held-out data. One fails on corpus composition: the state-versus-independent tone difference, which depends on outlets assigned to the Russian independent register under an incorrect application of the ownership rule, and which moves *further* from significance at each successive correction to that register, ending at $p = 0.561$.

Taken together these six bound how much confidence a single significant specification warrants in this literature. Each is significant at conventional levels, each admits a plausible economic mechanism, and each survives at least one robustness check before failing another. Two are eliminated by a better control, one by additional data, one by a rule fixed before the data existed, one by holding data back, and one by correcting a classification error — and in every case the eliminating test is one that a study reporting the positive would have had no particular reason to run. They are reported here because in a literature whose regressors are constructed from text, the set of choices under which a result survives is part of the result, and a null established against this background is considerably more informative than one obtained without it.

5 Conclusion

The Western narrative is what is priced in Western defence equities. Local-language perception of the Russia–Ukraine war — from Ukrainian outlets, Russian state media, and Russian independent outlets — adds no explanatory power to defence returns conditional on Western coverage, in any of four tests, across two asset classes, and against one non-market outcome. The null is not attributable to a design unable to detect media effects: the Western block is detected under correction in the anticipation and gas tests, and in the attention-and-tone test the local block itself is detected on the day of publication and lost when lagged by the single day required for tradeability. A power simulation on 1,855 out-of-sample days establishes that effects in the range the predictability literature treats as economically meaningful would have been detected in 83% to 100% of simulated samples. No such effect is observed.

This result is the mirror image of Bondarenko et al. (2024): their local-versus-English asymmetry runs in the opposite direction for the other side of the conflict. For the Russian domestic economy the local narrative dominates; for Western defence equities the Western narrative dominates. Neither result is a general claim about whether local media matter; together they say that each market prices the information ecosystem that is closest to its own participants.

The practical implication extends beyond this conflict. Multilingual geopolitical-risk indicators are constructed on the premise that sources closer to an event carry information Anglophone coverage omits. Tested in the setting most favourable to it, that premise does not hold: a Ukrainian- or Russian-language risk index constructed for the purpose of trading Western defence equities measures a signal those prices do not respond to. The premise is not refuted in general, since Bondarenko et al. (2024) establish it for the Russian domestic economy. What follows is that it cannot be assumed. Whether local-language sources add information is an empirical question about the distance between an information ecosystem and a market's marginal investor, and it must be settled market by market rather than in general.

Three limitations bound what this thesis can claim. The first is that the analysis is associational rather than causal: coverage and prices respond to the same underlying events, and the conditional design separates ecosystems from one another but does not identify a causal effect of coverage on prices. The second is measurement: GDELT tone is a dictionary-based score, and Loughran and McDonald (2011) give good reason to treat such scores as indicators of coverage negativity rather than of investor belief; multilingual transformer models applied to article text would sharpen the signal, though they would not change the classification question this thesis is about. The third is frequency: at daily resolution a reaction completed within the trading day is invisible, and the Gate 2 timing result — which turns on a one-day lag — is precisely the kind of finding that intraday data could either sharpen or overturn.

One apparent limitation can be discharged directly. An index-level null may in principle mask cross-sectional variation, with heavily war-exposed firms responding where a diversified index does not. A firm-level panel of 31 defence names and 85,065 firm-days, with exposure measured from published SIPRI arms-revenue data, produces no exposure gradient in any war window: nominal significance appears only in the pre-war period and in a full sample that is 59% pre-war, and none of the ten cells survives correction. The index-level result is therefore not an artefact of aggregation.

The extension of greatest value would establish whether this result generalises. Applying the same ecosystem comparison to other conflicts, such as the Hamas–Israel war or tensions in the Taiwan Strait, would distinguish a general property of defence-equity pricing from a feature specific to this war, in which the local-language ecosystems were already at coverage saturation before its defining event. A second extension would move to intraday prices, at which the anticipated-versus-realised distinction that Gate 3 cannot resolve at daily frequency becomes observable. A third would apply the design to assets whose marginal investors are not Western — rouble-denominated instruments, or the sovereign debt of states bordering the conflict — which constitutes the sharpest available test of the proposition that each market prices the information ecosystem its own participants inhabit.

References

Apergis, N., Bonato, M., Gupta, R., and Kyei, C. (2018). Does geopolitical risks predict stock returns and volatility of leading defense companies? evidence from a nonparametric approach. *Defence and Peace Economics*, 29(6):684–696.

Baker, S. R., Bloom, N., and Davis, S. J. (2016). Measuring economic policy uncertainty. *Quarterly Journal of Economics*, 131(4):1593–1636.

Benjamini, Y. and Hochberg, Y. (1995). Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B*, 57(1):289–300.

Bondarenko, Y., Lewis, V., Rottner, M., and Schüler, Y. (2024). Geopolitical risk perceptions. *Journal of International Economics*, 152:104005.

Caldara, D. and Iacoviello, M. (2022). Measuring geopolitical risk. *American Economic Review*, 112(4):1194–1225.

Campbell, J. Y. and Thompson, S. B. (2008). Predicting excess stock returns out of sample: Can anything beat the historical average? *Review of Financial Studies*, 21(4):1509–1531.

Clark, T. E. and West, K. D. (2007). Approximately normal tests for equal predictive accuracy in nested models. *Journal of Econometrics*, 138(1):291–311.

Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2):174–196.

Covachev, S. and Fazakas, G. (2024). The effects of the Russia–Ukraine war and the Wagner Group coup on defense stocks in Europe: An event study analysis. *Studies in Economics and Finance*, 42(1):31–42.

Diebold, F. X. and Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263.

Harvey, D., Leybourne, S., and Newbold, P. (1997). Testing the equality of prediction mean squared errors. *International Journal of Forecasting*, 13(2):281–291.

Klomp, J. (2025). The impact of the Hamas–Israel conflict on the U.S. defense industry stock market return. *PLOS ONE*, 20(2):e0314677.

Leetaru, K. and Schrodt, P. A. (2013). GDELT: Global data on events, location and tone, 1979–2012. In *Annual Meeting of the International Studies Association*.

Loughran, T. and McDonald, B. (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *Journal of Finance*, 66(1):35–65.

Manela, A. and Moreira, A. (2017). News implied volatility and disaster concerns. *Journal of Financial Economics*, 123(1):137–162.

Newey, W. K. and West, K. D. (1987). A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708.

Tetlock, P. C. (2007). Giving content to investor sentiment: The role of media in the stock market. *Journal of Finance*, 62(3):1139–1168.

Welch, I. and Goyal, A. (2008). A comprehensive look at the empirical performance of equity premium prediction. *Review of Financial Studies*, 21(4):1455–1508.

Zhang, Y., He, J., He, M., and Li, S. (2023). Geopolitical risk and stock market volatility: A global perspective. *Finance Research Letters*, 53:103620.